

Aditya Kumar Sethia

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EDUCATION

Bachelor of Technology (B.Tech), Artificial Intelligence & Data Science
ICFAI Tech (ICFAI Foundation for Higher Education, IFHE), Hyderabad, India
(Category I Autonomy, Deemed to be University)

CGPA: 9.77 / 10
Graduating: Summer 2027

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash

ML & Data: PyTorch, PyTorch Lightning, TensorFlow/Keras, LightGBM, Hugging Face / sentence-transformers, pandas, LangChain, ChromaDB

Backend & Tools: FastAPI, Flask, RESTful APIs, Git/GitHub, CI/CD, pytest, Linux (Ubuntu/WSL), CUDA, mixed-precision training

Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Software Engineering Intern (Python & AI) | Eficens Software Services June 2025 to July 2025

- Engineered a Retrieval Augmented Generation service behind a secure FastAPI REST backend, integrating LangChain, PyTorch, and ChromaDB to vectorize web content and serve LLM queries.
- Automated Ubuntu 24.04 security auditing through a suite of 300+ Bash scripts enforcing CIS benchmark controls, replacing manual per-host review with repeatable, consistent compliance checks.

OPEN SOURCE CONTRIBUTIONS

Kornia (Computer Vision Library) | Merged Contributor

- Vectorized RGB $\leftrightarrow$ YUV color conversion for a 5.70x CUDA speedup (19.7ms $\rightarrow$ 3.4ms) and 1.86x CPU speedup (1527ms $\rightarrow$ 820ms) under torch.compile (Inductor), verified numerically equivalent via pytest smoke, exception, and gradcheck suites.
- Reduced rgb_to_grayscale CUDA latency by 34% (477ms $\rightarrow$ 315ms) by fusing multiply-add operations with addcmul to eliminate intermediate memory allocations (merged PR #3474).
- Resolved a CPU/GPU device mismatch RuntimeError in the RandomGaussianIllumination augmentation, submitting a minimal reproduction script and crash log alongside the fix (merged PR #3457).

neural-lam (PyTorch Lightning) | Merged Contributor

- Restored single-GPU test suite execution (merged PR #193, closing issue #192) by replacing a Trainer configuration hardcoded to two devices with dynamic device detection.
- Traced the resulting crash to a tensor rank bug where PyTorch Lightning's all_gather returns 1D metrics on a single device; added dimension guards and cleared a deprecated transpose warning.

CNCF KubeEdge Ianvs | Code Review & Defect Analysis

- Identified a single root-cause defect pattern, unguarded division by a count that can legitimately be zero, recurring across 4 independent subsystems that had each been filed as an unrelated issue.
- Proved 3 concurrent open pull requests were patching identical source lines by reproducing every pairwise git merge conflict, then recommended a consolidation path to maintainers publicly.
- Reviewed 15 open issues and pull requests with executable reproduction commands, surfacing a previously unreported latent ZeroDivisionError by auditing all 21 repository configurations.

TECHNICAL PROJECTS

Amazon ML Challenge 2026: Multilingual Entity Resolution | Team Lead github.com/AdityaKumarSethia/amazon-ml-26

- Led a 3-person team building a blocking-and-matching pipeline that resolved $\sim$ 1.7M reference businesses against two noisy, multilingual sources with no shared identifiers, scoring 0.981 macro F0.5 on the public leaderboard.
- Designed hybrid candidate generation unioning sharded TF-IDF lexical retrieval, multilingual e5 dense embeddings (exact GPU matmul search), and deterministic keys on normalized names, street numbers and postcodes, keeping per-entity candidate sets small for the organizers' blocking-efficiency review.
- Trained a LightGBM pairwise matcher on string-similarity and rank-gap features, tuning the threshold on held-out F0.5 (0.989 validation), and shipped a reproducible package verified by hash to rebuild leaderboard outputs byte-identically.

Fluid Diffusion LAM: Generative PDE Surrogate | PyTorch Lightning github.com/AdityaKumarSethia/fluid-diffusion-lam

- Built a modular UNet latent diffusion surrogate emulating 2D PDE dynamics (Shallow Water and Reaction Diffusion systems) from PDEBench data, generating 5-member probabilistic ensembles scored on RMSE, CRPS, and Spread Skill Ratio.
- Designed an autoregressive rollout with a sliding-window history buffer and ensemble-mean stabilization to sustain multistep forecasts without error accumulation, trained under 16-bit mixed precision on a single consumer GPU.
- Built a preparation pipeline converting raw PDEBench HDF5 archives into normalized sequential tensors, feeding a rolling-window dataset and dataclass-driven configs so one training loop serves both single-channel fluid and two-channel chemical systems.

MobileNetV2 From Scratch: Plant Disease Classifier | TensorFlow, Keras github.com/AdityaKumarSethia/PlantDiseaseDetection

- Implemented MobileNetV2 from the original paper in the Keras Functional API (inverted residuals, linear bottlenecks) without pretrained weights, reaching 98.4% validation accuracy across 38 classes on 87,000+ images.
- Built a tf.data pipeline with caching, prefetching, and GPU-resident augmentation, and enabled mixed_float16 Tensor Core training to complete 48 epochs in 8.5 hours on a single 8 GB GPU.
- Packaged a standalone evaluation script that runs batch inference on unlabelled images and emits a confusion matrix, per-class report, and annotated prediction grid with confidence scores.